

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Mid-Semester Test
EC-2 Regular Test Solution

Course No. : DSECL ZC416/AIMLC ZC416
Course Title : Mathematical Foundations for Machine Learning/Data Science
Nature of Exam : Closed Book
Weightage : 30%

Q1. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & -1 & 0 \end{bmatrix}$.

(a) Solve the system of equations $AX = b$, where $b = \begin{bmatrix} 6 \\ 12 \\ 5 \end{bmatrix}$ by finding the Echelon form of the augmented matrix. [3M]

$$(a) \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 4 & 6 & 12 \\ 1 & -1 & 0 & 5 \end{array} \right] \approx \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & -3 & -1 \end{array} \right] \approx \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -3 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad [2M]$$

Converting back to equation

$$a + 2b + 3c = 6$$

$$-3b - 3c = -1$$

Since 3rd column in the augmented matrix has no pivot element the system has infinitely many solutions. Take $c = k$, we get $b = \frac{1}{3}[1 - 3k]$, and $a = 6 - \frac{2}{3}[1 - 3k] - k = \frac{16}{3} + k$. [1M]

The complete set of solutions is $\left\{ \begin{bmatrix} \frac{16}{3} + k \\ \frac{1}{3}[1 - 3k] \\ k \end{bmatrix} : k \text{ is a real number} \right\}$

(b) Find the Echelon form of $B = \begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 5 & -3 \end{bmatrix}$. Hence find the determinant of B . Also find the rank of B . [2M]

$$\begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 5 & -3 \end{bmatrix} \approx \begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 0 & -5 & 9 \\ 0 & 0 & 0 & 6 \end{bmatrix} \quad [1M]$$

$$\text{Det } B = 5 * (-1) * (-5) * 6 = 150. \quad [0.5M]$$

Since there are four non-zero rows in the ERF, rank of B is 4.

Q2. Consider the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ in R^3 .

(a) Write $\begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ as linear combination of the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}$. Explain how this will imply the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ are linearly dependent in R^3 . [3M]

Solution:

$$\text{Suppose } a \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + b \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$$

We have to solve the system of linear equations whose augmented matrix is given by

$$\begin{bmatrix} 5 & 6 & 4 \\ -3 & 2 & -8 \\ 1 & -3 & 5 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 5 \\ -3 & 2 & -8 \\ 5 & 6 & 4 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 5 \\ 0 & -7 & 7 \\ 0 & 21 & -21 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 5 \\ 0 & -7 & 7 \\ 0 & 0 & 0 \end{bmatrix} \quad [1M]$$

Converting back to equations:

$$a - 3b = 5$$

$$-7b = 7 \text{ implies } b = -1$$

$$\text{Hence } a = 5 + 3b = 5 - 3 = 2.$$

$$\text{Therefore } 2 \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix} \quad [1M]$$

We rewrite the above equation as

$$2 \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} + (1) \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This implies that vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ are linearly dependent in R^3 . [1M]

(b) Prove that the $V = \text{span} \left\{ \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} \right\}$ is a subspace of R^3 . Find a basis and dimension of V . [2M]

In order to prove V is a subspace, It is enough to prove that V is closed with respect to vector addition and scalar multiplication.

Suppose $u, v \in V$. Then $u = a \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + b \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}$ and

$v = c \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + d \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}$ for some scalars a, b, c, d .

Then $u + v = (a + c) \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + (b + d) \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} \in V$ and what we

For any scalar e , $eu = u = ea \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix} + eb \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} \in V$. [1M]

If they simply quote the theorem that span of vectors is a subspace, please award full mark.

From what we have calculated in (a) the echelon form of the matrix $\begin{bmatrix} 5 & 6 \\ -3 & 2 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 0 & -7 \\ 0 & 0 \end{bmatrix}$.

Since each column has pivot element, $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}$ are linearly independent.

Since already they span V , $\left\{ \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} \right\}$ is a basis of V and the dimension of V is 2. [1M]

(c) Is the element $\begin{bmatrix} 4 \\ -8 \\ 2 \end{bmatrix}$ in V where V is the subspace defined in (b). Justify your answer. [1M]

$$\begin{bmatrix} 5 & 6 & 4 \\ -3 & 2 & -8 \\ 1 & -3 & 2 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 2 \\ -3 & 2 & -8 \\ 5 & 6 & 4 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 2 \\ 0 & -7 & -2 \\ 0 & 21 & -21 \end{bmatrix} \approx \begin{bmatrix} 1 & -3 & 2 \\ 0 & -7 & 7 \\ 0 & 0 & -27 \end{bmatrix}$$

Converting back to equation the last row we get $0 = -27$. Hence $\begin{bmatrix} 4 \\ -8 \\ 2 \end{bmatrix}$ is not in V . [1M]

Q3. (a) Find the Singular Value decomposition of $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$. [4M]

(b) Is the matrix $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ diagonalizable? Justify your answer. [2M]

Singular Value Decomposition and Diagonalizability

(a) Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

First, compute $A^T A$ and AA^T to find the singular values.

$$A^T A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Eigenvalues of $A^T A$ (or AA^T) are the same (nonzero ones).
Characteristic equation for $A^T A$:

$$\det(A^T A - \lambda I) = \det \begin{pmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4) = 0$$

$$\implies \lambda_1 = 4, \lambda_2 = 0$$

Singular values: $\sigma_1 = \sqrt{4} = 2, \sigma_2 = 0$

Now find the SVD: $A = U\Sigma V^T$

Step 1: Right singular vectors (eigenvectors of $A^T A$)

For $\lambda_1 = 4$:

$$(A^T A - 4I)v_1 = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix} v_1 = 0$$

$$\implies -2x + 2y = 0 \implies x = y$$

Unit vector: $v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

For $\lambda_2 = 0$:

$$(A^T A)v_2 = 0 \implies \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x + y = 0$$

Unit vector: $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Step 2: Left singular vectors (from $u_i = \frac{1}{\sigma_i} A v_i$ for $\sigma_i > 0$)

For $i = 1$:

$$A v_1 = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ -2 \end{pmatrix} = \sqrt{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{2} \cdot \sqrt{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

For $i = 2$, we can choose any unit vector orthogonal to u_1 , e.g.

$$u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

1

Step 3: Singular value matrix

$$\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$$

Thus, one possible SVD is:

$$A = \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}}_U \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^T}_{V^T}$$

(Note: signs in U and V may differ by ± 1 depending on choice.)

(b) Is the matrix $A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ diagonalizable? Justify your answer.

To check diagonalizability over the real numbers, compute the eigenvalues of A .
Characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ -1 & -1-\lambda \end{pmatrix} = (1-\lambda)(-1-\lambda) - (1)(-1) = -1 - \lambda + \lambda + \lambda^2 + 1 = \lambda^2$$

$$\det(A - \lambda I) = \lambda^2 \implies \lambda = 0 \text{ (algebraic multiplicity 2)}$$

Now find eigenvectors:

$$(A - 0I)v = Av = 0 \implies \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$x + y = 0 \implies y = -x$$

So eigenvectors are multiples of $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

The eigenspace has dimension 1 (geometric multiplicity = 1).

Since algebraic multiplicity (2) > geometric multiplicity (1), the matrix A is **not diagonalizable** over $\mathbb{R}$.

No, A is not diagonalizable.

Marking Scheme:

1. The singular values are found from eigenvalues of $A^T A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$. The eigenvalues are 4 and 0, so $\sigma_1 = 2$, $\sigma_2 = 0$.
2. The right singular vectors (columns of V) are eigenvectors of $A^T A$: $v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.
3. The left singular vectors (columns of U) are obtained from $u_1 = \frac{Av_1}{\sigma_1}$ and u_2 orthogonal to u_1 , giving $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.
4. The SVD is $A = U\Sigma V^T$, where $\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ contains the singular values in descending order.
5. The characteristic polynomial is $\det(A - \lambda I) = \lambda^2 = 0$, so $\lambda = 0$ is the only eigenvalue with algebraic multiplicity 2.
6. Eigenvectors satisfy $A\mathbf{v} = 0$, giving solutions of the form $\begin{pmatrix} t \\ -t \end{pmatrix}$. The eigenspace has dimension 1, so A is not diagonalizable.

Each point carries 1 Mark.

Q4. Consider the matrix $A = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$.

(a) Define an inner product on R^2 using the matrix A by $\langle X, Y \rangle = X^T A Y$. What are the properties of A that ensures that $\langle X, Y \rangle = X^T A Y$ is an inner product. Prove those properties of A . [3M]

Solution:

Clearly $A = A^T$ Hence A is symmetric. [1M]

For a symmetric matrix, A is positive definite iff all leading principal minors are positive.

For A :

$$\det(3) = 3 > 0, \det(A) = \begin{vmatrix} 3 & -1 \\ -1 & 1 \end{vmatrix} = 3 - 1 = 2 > 0.$$

Hence A is **positive definite**. [2M]

If students prove $X^T A Y > 0$ for any non zero X in R^2 please award full mark.

Therefore, $\langle X, Y \rangle = X^T A Y$ is an inner product since A is symmetric and positive definite.

(b) Using this inner product find the distance between the vectors $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. [2M]

Given the inner product

$$\langle X, Y \rangle = X^T A Y, A = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix},$$

the distance between two vectors u and v is

$$d(u, v) = \sqrt{\langle u - v, u - v \rangle}.$$

$$\text{Let } u = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\text{Then } u - v = \begin{pmatrix} -1 \\ -2 \end{pmatrix}. \quad [1/2 \text{ M}]$$

$$\langle u - v, u - v \rangle = (u - v)^T A (u - v)$$

$$A(u - v) = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad [1/2 \text{ M}]$$

$$(u - v)^T A (u - v) = (-1 \quad -2) \begin{pmatrix} -1 \\ -1 \end{pmatrix} = 1 + 2 = 3$$

$$d(u, v) = \sqrt{3}. \quad [1\text{M}]$$

(c) Verify whether the vectors $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are orthogonal with respect to the inner product defined in (a)

Let

$$u = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$Av = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 - 1 \\ -2 + 1 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}.$$

$$\langle u, v \rangle = u^T Av = (-1 \quad 2) \begin{pmatrix} 5 \\ -1 \end{pmatrix} = -5 - 2 = -7.$$

$$\langle u, v \rangle = -7 \neq 0. \quad [1/2\text{M}]$$

Hence u and v are not orthogonal. [1/2M]

Q5. Consider the function

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3.$$

(a) Compute the gradient $\nabla f(x, y)$. [2M]

$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 4y \end{pmatrix}. \text{ [2M] If partly correct 1M.}$$

(b) Using (a) find the critical point(s) of f . [1M]

Critical points occur where the gradient is zero.

From part (a),

$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 4y \end{pmatrix}.$$

Set each component equal to zero:

$$\begin{cases} 2x + y - 4 = 0, \\ x + 4y = 0. \end{cases} \text{ [1/2M]}$$

From the second equation, $x = -4y$.

Substitute into the first:

$$2(-4y) + y - 4 = 0 \Rightarrow -8y + y - 4 = 0 \Rightarrow -7y = 4 \Rightarrow y = -\frac{4}{7}.$$

Then

$$x = -4y = \frac{16}{7}.$$

Critical point

$$\left(\frac{16}{7}, -\frac{4}{7} \right) \text{ [1/2 M]}$$

(c) Determine whether the critical point is a point of local minimum, local maximum, or saddle point.

To classify the critical point, use the **second derivative (Hessian) test**.

Given

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3,$$

Step 1: Second partial derivatives

$$f_{xx} = 2, f_{yy} = 4, f_{xy} = f_{yx} = 1.$$

So the Hessian matrix is

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}. \quad [1/2 \text{ M}]$$

Step 2: Determinant of the Hessian

$$D = \det(H) = (2)(4) - (1)^2 = 8 - 1 = 7 > 0. \quad [1/2 \text{ M}]$$

Step 3: Apply the test

- $D > 0$ and $f_{xx} = 2 > 0$
 $\Rightarrow$ the critical point is a **local minimum**.

Conclusion

The critical point $\left(\frac{16}{7}, -\frac{4}{7}\right)$ is a point of local minimum. [1M]

(d)

Let $g(x) = f(x, 1)$. Find the Taylor series expansion of $g(x)$ at $x = 1$. [2M]

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3$$

$$g(x) = x^2 + x(1) + 2(1)^2 - 4x + 3 = x^2 - 3x + 5$$

$$g'(x) = 2x - 3, g''(x) = 2, g^{(3)}(x) = 0$$

Evaluate at $x = 1$:

$$g(1) = 1 - 3 + 5 = 3,$$

$$g'(1) = 2 - 3 = -1$$

$$g''(1) = 2 \quad [1/2 \text{ M}]$$

$$g(x) = g(1) + g'(1)(x - 1) + \frac{g''(1)}{2!}(x - 1)^2 + \frac{g^{(3)}(1)}{2!}(x - 1)^2 + \frac{g^{(3)}(1)}{2!}(x - 1)^3 + \dots [1/2 \text{ M}]$$

Hence the Taylor series of $g(x)$ at $x = 1$:

$$g(x) = 3 - (x - 1) + (x - 1)^2. \quad [1M]$$

Q5. Consider the function

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3.$$

(a) Compute the gradient $\nabla f(x, y)$. [2M]

$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 4y \end{pmatrix}. \text{ [2M] If partly correct 1M.}$$

(b) Using (a) find the critical point(s) of f . [1M]

Critical points occur where the gradient is zero.

From part (a),

$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 4y \end{pmatrix}.$$

Set each component equal to zero:

$$\begin{cases} 2x + y - 4 = 0, \\ x + 4y = 0. \end{cases} \text{ [1/2M]}$$

From the second equation, $x = -4y$.

Substitute into the first:

$$2(-4y) + y - 4 = 0 \Rightarrow -8y + y - 4 = 0 \Rightarrow -7y = 4 \Rightarrow y = -\frac{4}{7}.$$

Then

$$x = -4 \left(-\frac{4}{7} \right) = \frac{16}{7}.$$

Critical point

$$\boxed{\left(\frac{16}{7}, -\frac{4}{7} \right)} \text{ [1/2 M]}$$

(c) Determine whether the critical point is a point of local minimum, local maximum, or saddle point.

To classify the critical point, use the **second derivative (Hessian) test**.

Given

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3,$$

Step 1: Second partial derivatives

$$f_{xx} = 2, f_{yy} = 4, f_{xy} = f_{yx} = 1.$$

So the Hessian matrix is

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}. \quad [1/2 \text{ M}]$$

Step 2: Determinant of the Hessian

$$D = \det(H) = (2)(4) - (1)^2 = 8 - 1 = 7 > 0. \quad [1/2 \text{ M}]$$

Step 3: Apply the test

- $D > 0$ and $f_{xx} = 2 > 0$
 $\Rightarrow$ the critical point is a **local minimum**.

Conclusion

The critical point $\left(\frac{16}{7}, -\frac{4}{7}\right)$ is a point of local minimum. [1M]

(d)

(d) Let $g(x) = f(x, 1)$. Find the Taylor series expansion of $g(x)$ at $x = 1$. [2M]

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3$$

$$g(x) = x^2 + x(1) + 2(1)^2 - 4x + 3 = x^2 - 3x + 5$$

$$g'(x) = 2x - 3, g''(x) = 2, g^{(3)}(x) = 0$$

Evaluate at $x = 1$:

$$g(1) = 1 - 3 + 5 = 3,$$

$$g'(1) = 2 - 3 = -1$$

$$g''(1) = 2 \quad [1/2 \text{ M}]$$

$$g(x) = g(1) + g'(1)(x - 1) + \frac{g''(1)}{2!}(x - 1)^2 + \frac{g^{(3)}(1)}{3!}(x - 1)^3 + \dots [1/2 \text{ M}]$$

Hence the Taylor series of $g(x)$ at $x = 1$:

$$g(x) = 3 - (x - 1) + (x - 1)^2. \quad [1\text{M}]$$
